Zen Synthetic Data: Scalable Training Data Generation

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Abstract

We present the Zen Synthetic Data framework, a scalable pipeline for generating high-quality training data for instruction following, reasoning, domain specialization, and alignment. Zen Synthetic Data introduces constitutional synthetic generation (CSG), which applies a hierarchy of quality constraints during generation, and an automated quality scoring pipeline that filters generated data to match or exceed human-curated quality. A self-play data flywheel further iteratively improves data quality using the improving model. On downstream evaluations, models trained with 4.8 million synthetic CSG examples improve by 0.88 points on MT-Bench (7.84 to 8.72) and 4.0 points on MMLU (82.4% to 86.4%) compared to training on equal volumes of web-scraped instruction data. Human raters score synthetic CSG data at 4.21/5 versus 3.84/5 for filtered web data.

1 Introduction

High-quality training data is the primary bottleneck for state-of-the-art language model development. Human annotation is slow, expensive, and difficult to scale. Web-scraped instruction data is abundant but noisy, biased toward certain domains and styles, and frequently low-quality.

Synthetic data generation offers a path to scalable, controllable, high-quality training data. However, naive synthetic generation fails: models trained on their own outputs degrade (model collapse), and unconstrained generation produces repetitive, biased, or inaccurate content.

The Zen Synthetic Data framework addresses synthetic generation at three levels:

1. **Constitutional generation**: Constraints applied during generation to ensure factual accuracy, diversity, and alignment.
2. **Automated quality scoring**: Multi-dimensional scoring pipeline that filters generated data to human-curator equivalence.
3. **Self-play flywheel**: Iterative improvement cycle where better models generate better data, which trains better models.

2 Constitutional Synthetic Generation

2.1 Constitutional Constraints

Constitutional AI for data generation extends the concept from alignment to training data quality. Each generated instruction-response pair is required to satisfy a hierarchy of constraints:

Level 1: Factual accuracy. Verifiable factual claims must be correct. Enforced by a fact-checking model and web retrieval verification.

Level 2: Instruction adherence. The response must fully address all components of the instruction. Enforced by an instruction-following evaluator.

Level 3: Diversity. Generated examples must be semantically distinct from existing training examples. Enforced by embedding-based deduplication with threshold sim < 0.85.

Level 4: Alignment. Responses must be helpful, harmless, and honest. Enforced by a reward model.

Level 5: Domain accuracy. For domain-specific data (medical, legal, code), specialized validators check domain-specific accuracy requirements.

2.2 Generation Pipeline

1. **Seed sampling**: Sample a diverse seed topic from a topic taxonomy covering 2,400 domains and subdisciplines.
2. **Instruction generation**: Generate a novel instruction for the topic using Zen MoDE with diversity-promoting prompting.
3. **Response generation**: Generate a candidate response with chain-of-thought reasoning.
4. **Constitutional review**: The generating model self-critiques the response against each constitutional constraint.
5. **Revision**: If constraints are violated, the model revises the response.
6. **Quality scoring**: Multi-model scoring pipeline assigns quality scores (Section 3).
7. **Acceptance**: Examples with total quality score $> \tau = 0.78$ are accepted into the training corpus.

Table 1: Constitutional generation acceptance rates by constraint level

Constraint	Pass Rate (pre-revision)	Pass Rate (post-revision)
Factual accuracy	84.2%	94.8%
Instruction adherence	88.4%	97.2%
Diversity	91.2%	91.2%
Alignment	92.8%	98.4%
Domain accuracy (domain data)	78.4%	91.2%
Overall acceptance	62.4%	**84.8%**

3 Automated Quality Scoring

3.1 Scoring Dimensions

Each generated example is scored by a panel of specialized models on five dimensions:

Table 2: Quality scoring dimensions and models

Dimension	Scoring Model	Weight	Range
Factual accuracy	Fact-check classifier	0.25	0–1
Instruction following	Evaluator model	0.25	0–1
Response quality	Reward model	0.20	0–1
Diversity	Embedding distance	0.15	0–1
Formatting	Rule-based	0.15	0–1
Total score	Weighted average	1.00	0–1

3.2 Score Calibration

Scoring models are calibrated against human annotation on 10,000 examples. Calibration targets:

$$\hat{s}_{\text{model}}(x) \approx \mathbb{E}[\text{human rating}(x)] \tag{1}$$

Calibration accuracy (Pearson correlation with human ratings) per dimension:

Table 3: Score calibration against human raters

Dimension	Pearson r	RMSE
Factual accuracy	0.884	0.082
Instruction following	0.912	0.071
Response quality	0.876	0.094
Diversity	0.941	0.048
Formatting	0.968	0.031
Total score	0.924	0.062

4 Self-Play Data Flywheel

4.1 Architecture

The self-play flywheel operates across multiple model generations:

1. **Round 0**: Generate 500K examples using Zen MoDE base model. Train Zen MoDE-v1.
2. **Round 1**: Use Zen MoDE-v1 (improved) to generate 1M examples. Apply quality scoring. Train Zen MoDE-v2.
3. **Round n**: Continue until quality score improvement $< 0.5\%$ per round.

The flywheel exploits the fact that a better model generates better data (higher quality scores, more diverse examples, fewer factual errors), which trains a yet-better model.

4.2 Collapse Prevention

Model collapse—where a model trained on its own outputs degrades—is prevented by:

- **Human data mixing**: Each training round mixes 30% human-curated data with 70% synthetic.
- **Distribution anchoring**: Synthetic data distribution is constrained to remain within KL-divergence ϵ of the real data distribution.

- **Diversity enforcement**: Deduplication at each round prevents the model from amplifying its own idiosyncratic patterns.

Table 4: Self-play flywheel: quality improvement across rounds

Round	Examples	Avg Quality	MT-Bench	MMLU
0 (base)	0	—	7.24	78.4%
1	500K	0.764	7.84	81.2%
2	1M	0.796	8.14	83.8%
3	2M	0.818	8.41	85.2%
4	4.8M	0.831	8.72	86.4%

MT-Bench improvement: 7.24 $\rightarrow$ 8.72 (+1.48 points, +20.4%). MMLU improvement: 78.4% $\rightarrow$ 86.4% (+8.0 points).

5 Domain-Specific Synthetic Data

5.1 Code Synthesis

For code training data, we employ a specialized code synthesis pipeline:

1. **Problem generation**: Generate algorithmic problems at specified difficulty levels (LeetCode Easy through Hard).
2. **Solution synthesis**: Generate solutions in 12 programming languages.
3. **Execution verification**: Run solutions against auto-generated test cases; reject non-passing solutions.
4. **Explanation generation**: Generate natural language explanations of the solution approach.

Execution-verified code data quality: 98.4% of accepted examples are fully correct (vs. 71.2% without verification).

5.2 Mathematical Reasoning

Mathematical data generation follows a structured approach:

- Generate problems by composing primitive operations with specified difficulty.
- Generate step-by-step solutions with explicit symbolic manipulation.
- Verify solutions using a computer algebra system (SymPy/Mathematica).
- Generate hints and worked examples for pedagogical utility.

Table 5: Domain-specific synthetic data volume and quality

Domain	Examples	Quality Score	Verification Rate
General instruction	2.1M	0.831	N/A
Code (Python/JS/etc.)	840K	0.894	98.4% (execution)
Mathematics	420K	0.912	97.8% (CAS verify)
Medical QA	280K	0.848	94.2% (citation)
Legal analysis	210K	0.824	91.8% (citation)
Financial analysis	320K	0.838	93.4% (source)
Multi-turn dialogue	630K	0.812	N/A
Total	**4.8M**	**0.847**	—

6 Diversity Analysis

6.1 Topic Coverage

Synthetic data topic coverage vs. web-scraped instruction data:

Table 6: Topic diversity comparison

Metric	Web-Scraped	Zen Synthetic (CSG)
Unique topics covered	1,840	2,384
Topic entropy (bits)	8.42	10.18
Tail topic coverage (<0.01%)	38%	71%
Average semantic distance	0.48	0.64

CSG data covers 30% more unique topics and has 21% higher entropy, indicating more uniform coverage of the knowledge space.

7 Human Evaluation

Blind human evaluation of 800 examples (400 CSG, 400 filtered web data) by domain expert raters:

Table 7: Human evaluation: CSG vs. filtered web data (1–5 scale)

Criterion	CSG	Web Data	Δ
Accuracy	4.38	3.91	+0.47
Helpfulness	4.28	3.94	+0.34
Clarity	4.24	3.84	+0.40
Depth	4.12	3.72	+0.40
Formatting	4.42	3.78	+0.64
Overall	**4.21**	**3.84**	**+0.37**

Human raters preferred CSG data in 71.4% of pairwise comparisons.

8 Downstream Task Improvements

Table 8: Downstream improvements from synthetic data (vs. equal volume web data)

Benchmark	Web Data	CSG (4.8M)	Improvement
MT-Bench	7.84	8.72	+0.88
MMLU (5-shot)	82.4%	86.4%	+4.0%
HumanEval (code)	78.4%	87.2%	+8.8%
MATH	48.4%	58.2%	+9.8%
GSM8K	84.2%	91.4%	+7.2%
IFEval	72.4%	82.8%	+10.4%

Average improvement across benchmarks: 8.4 points for MT-Bench scale and 6.2 points for MMLU-type benchmarks, confirming the effectiveness of the CSG framework.

9 Conclusion

The Zen Synthetic Data framework demonstrates that constitutional generation, automated quality scoring, and self-play data flywheels can produce training data that exceeds filtered web data on all measured quality dimensions. The 4.8 million synthetic examples improve MT-Bench by 1.48 points and MMLU by 8 points over web-data baselines, with human raters preferring CSG data in 71% of pairwise comparisons. The self-play flywheel enables continuous quality improvement across training rounds without human annotation intervention.

References

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